

Chp21 极限

在初中，我们学过半径为 r 的圆的面积为 $A = \pi r^2$ 。
同学们记得这个公式是怎样推导出来的吗？



图1

圆分成若干等份

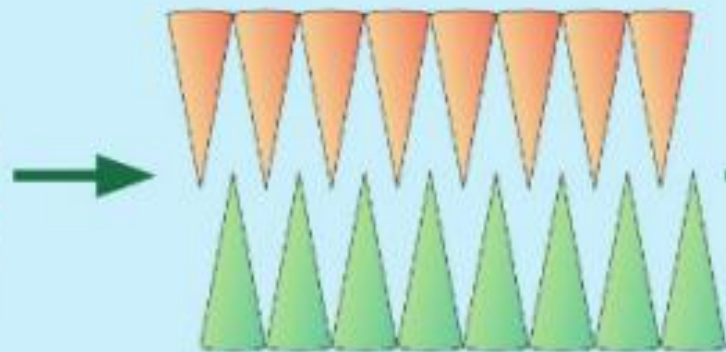


图2

剪开，如图上下排列

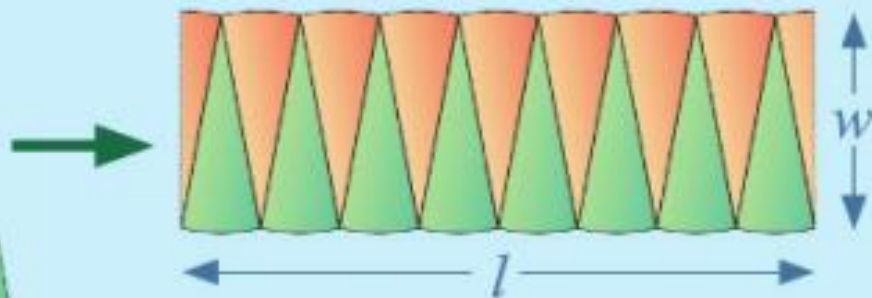


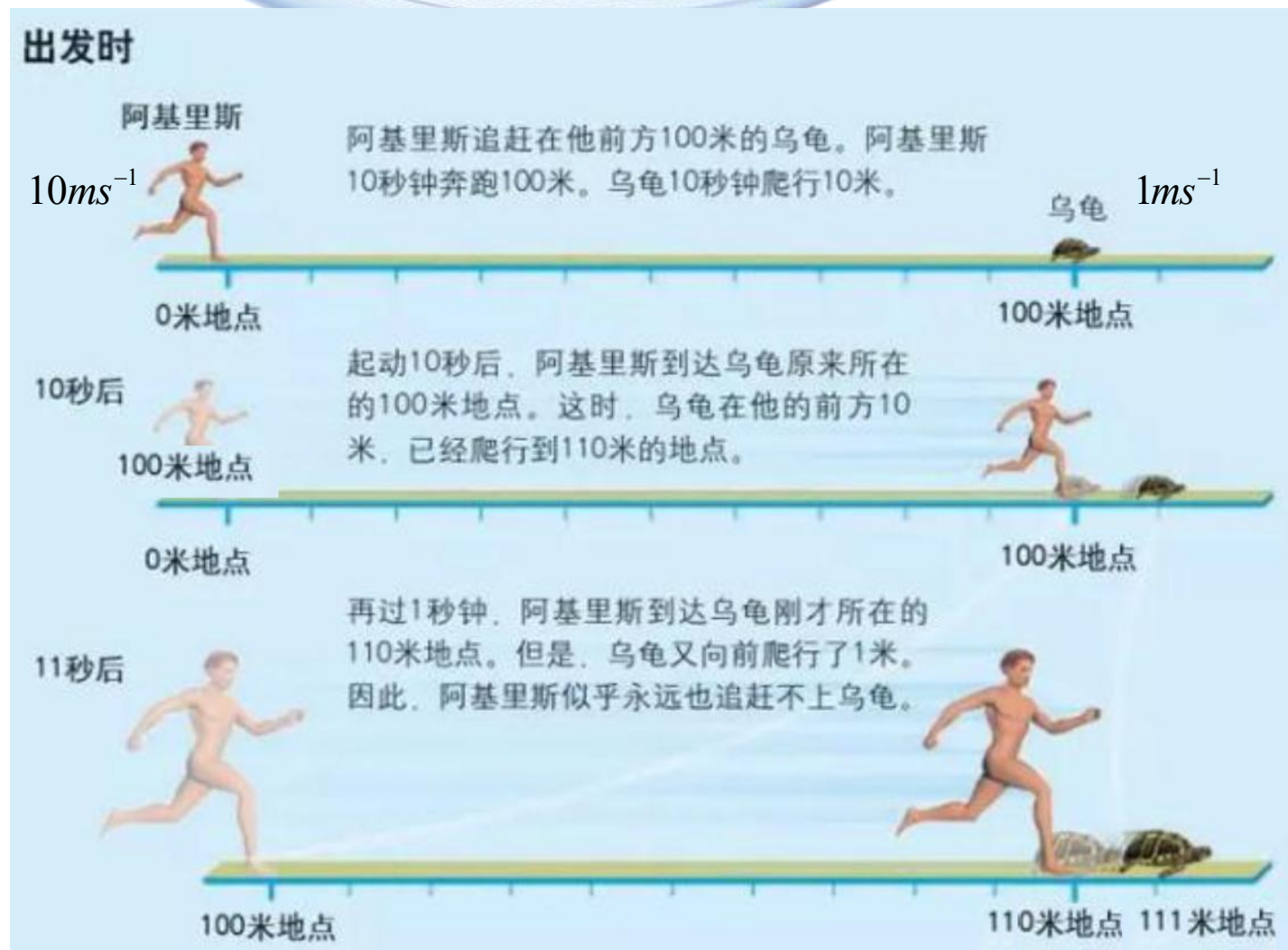
图3

如图拼起来

割圆术



芝诺悖论（第二次数学危机）



<https://www.youtube.com/watch?v=a4J7yXPksOA>



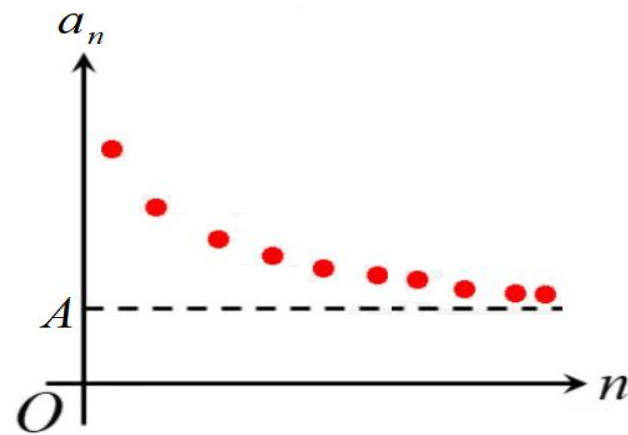
芝诺的乌龟（下）



21.1 数列的极限

存在一个常数 A ,使得数列 $\{a_n\}$ 当 $n \rightarrow \infty, a_n \rightarrow A$,
则称数列 $\{a_n\}$ 的极限(*limit*)为 A ,
或说数列 $\{a_n\}$ 收敛(*converge*)至 A

记作 $\lim_{n \rightarrow \infty} a_n = A$



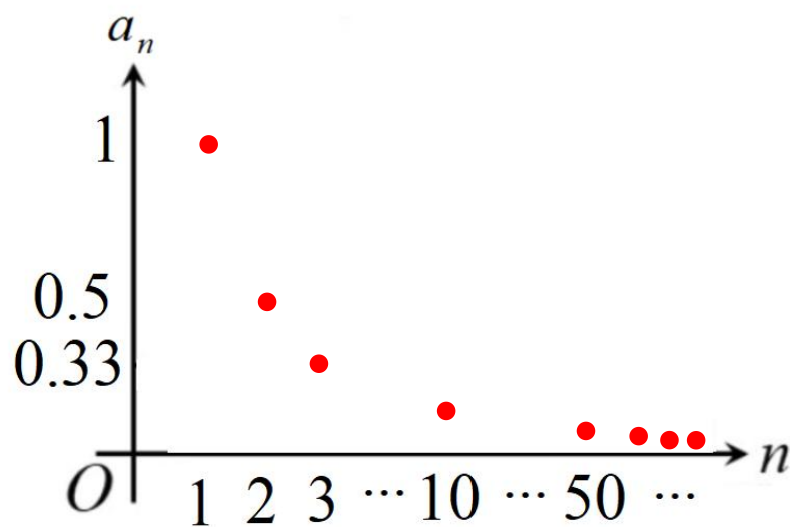
注:" $\rightarrow$ "意指"趋近"

" ∞ "意指无限大或无穷大(*infinity*)

"lim"意指*limit*的简写

a)求数列 $\{a_n\} = \frac{1}{n}$ 的极限。

n	1	2	3	10	50	100	1000	...
a_n	1	0.5	0.33	0.1	0.02	0.01	0.001	...

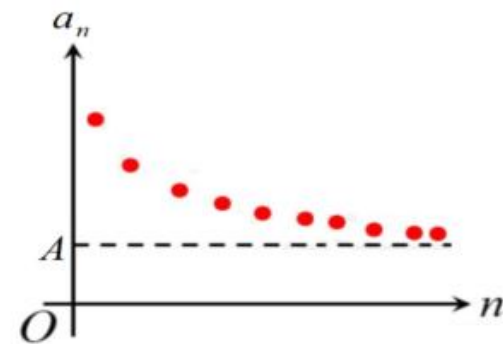


$$\therefore \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

结论: $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

存在一个常数 A ,使得数列 $\{a_n\}$ 当 $n \rightarrow \infty, a_n \rightarrow A$,
则称数列 $\{a_n\}$ 的极限(*limit*)为 A ,
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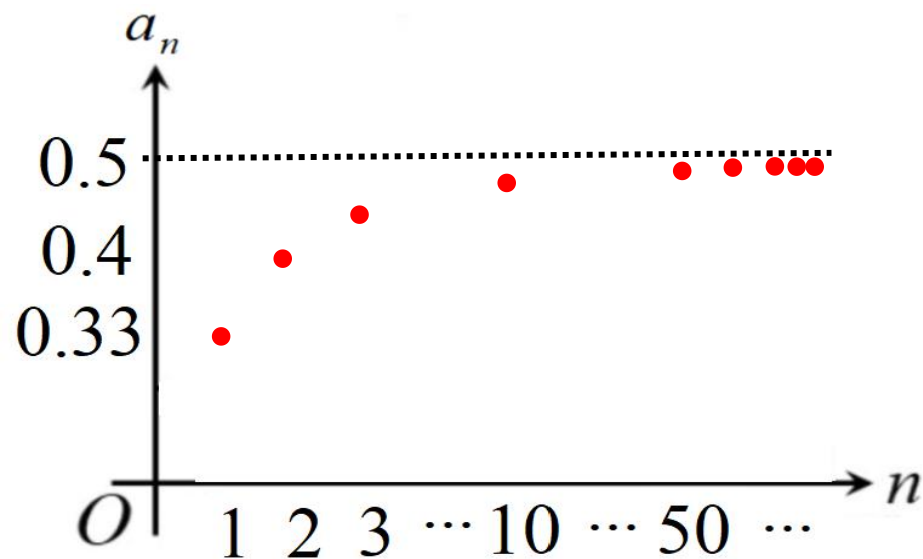
"lim"意指*limit*的简写

b) 求数列 $\{a_n\} = \frac{n}{1+2n}$ 的极限。

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

解一

n	1	2	3	10	50	100	1000	...
a_n	0.33	0.4	0.429	0.476	0.495	0.498	0.4998	...



$$\therefore \lim_{n \rightarrow \infty} \frac{n}{1+2n} = \frac{1}{2}$$

解二 $\therefore \lim_{n \rightarrow \infty} \frac{n}{1+2n} = \lim_{n \rightarrow \infty} \frac{\frac{n}{n}}{\frac{1}{n} + \frac{2n}{n}} = \lim_{n \rightarrow \infty} \frac{\frac{n}{n}}{\frac{1}{n} + 2} = \frac{1}{2}$

c) 求数列 $\{a_n\} = \frac{2n^2 - 1}{n^2 + 2n + 3}$ 的极限。

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{2n^2 - 1}{n^2 + 2n + 3}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{2n^2 - 1}{n^2}}{\frac{n^2 + 2n + 3}{n^2}}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{2n^2}{n^2} - \frac{1}{n^2}}{\frac{n^2}{n^2} + \frac{2n}{n^2} + \frac{3}{n^2}}$$

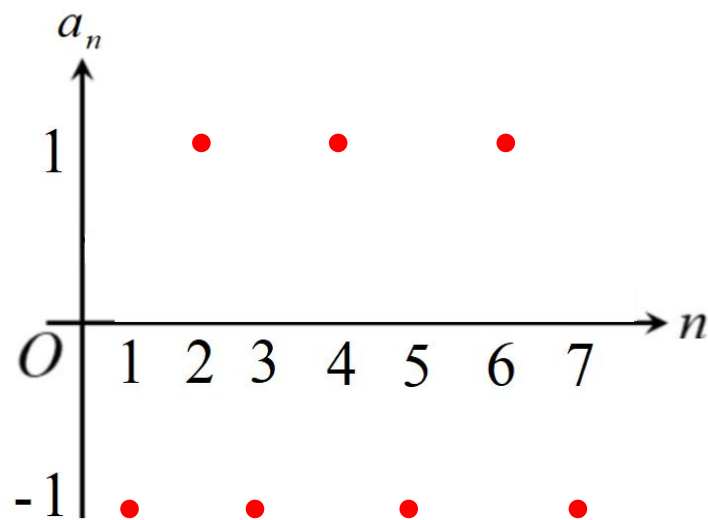
$$= \lim_{n \rightarrow \infty} \frac{2 - \frac{1}{n^2}}{1 + \frac{2}{n} + \frac{3}{n^2}}$$

$$= \frac{2 - 0}{1 + 0 + 0}$$

$$= 2$$

d)求数列 $\{a_n\} = (-1)^n$ 的极限。

n	1	2	3	4	5	6	7	...
a_n	-1	1	-1	1	-1	1	-1	...



$\therefore \lim_{n \rightarrow \infty} (-1)^n = \text{不存在}$

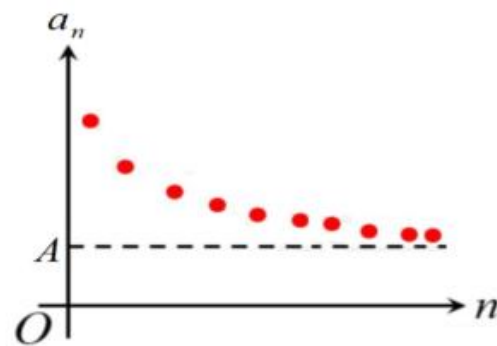
存在一个常数 A ,使得数列 $\{a_n\}$ 当 $n \rightarrow \infty, a_n \rightarrow A$,
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或说数列 $\{a_n\}$ 收敛(*converge*)至 A

记作 $\lim_{n \rightarrow \infty} a_n = A$

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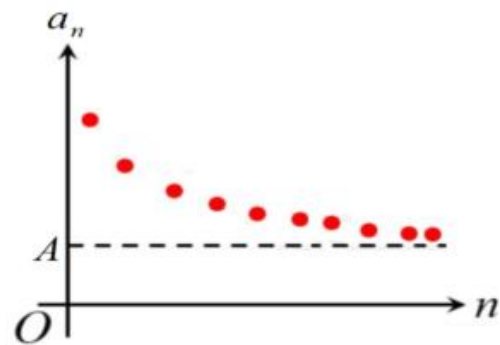


e)求数列 $\{a_n\}$ = 2025的极限。

n										...
a_n										...

存在一个常数 A ,使得数列 $\{a_n\}$ 当 $n \rightarrow \infty, a_n \rightarrow A$,
则称数列 $\{a_n\}$ 的极限(*limit*)为 A ,
或说数列 $\{a_n\}$ 收敛(*converge*)至 A

记作 $\lim_{n \rightarrow \infty} a_n = A$



注:" $\rightarrow$ "意指"趋近"

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"lim"意指*limit*的简写

$\lim_{n \rightarrow \infty} k = k$, k 为常数

$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

f) 已知 $\lim_{n \rightarrow \infty} a_n = 5, \lim_{n \rightarrow \infty} b_n = 3$, 求 $\lim_{n \rightarrow \infty} \left(\frac{a_n + 5b_n}{7a_n} \right)$ 的值。

数列的极限

若 $\lim_{n \rightarrow \infty} a_n$ 与 $\lim_{n \rightarrow \infty} b_n$ 存在, k 为常数

$$i) \lim_{n \rightarrow \infty} (ka_n) = k \lim_{n \rightarrow \infty} a_n$$

$$ii) \lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$$

$$iii) \lim_{n \rightarrow \infty} (a_n \cdot b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$$

$$iv) \lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}, \text{ 其中 } \lim_{n \rightarrow \infty} b_n \neq 0$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{a_n + 5b_n}{7a_n} \right) &= \frac{\lim_{n \rightarrow \infty} (a_n + 5b_n)}{\lim_{n \rightarrow \infty} 7a_n} \\ &= \frac{\lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} 5b_n}{\lim_{n \rightarrow \infty} 7a_n} \\ &= \frac{\lim_{n \rightarrow \infty} a_n + 5 \lim_{n \rightarrow \infty} b_n}{7 \lim_{n \rightarrow \infty} a_n} \\ &= \frac{5 + 5(3)}{7(5)} \\ &= \frac{4}{7} \end{aligned}$$

当 $n \rightarrow \infty$, 分子、分母同时无限增大,
 则须将分子、分母同除以 n 的最高次方后再求极限。

g)求下列的极限:

$$i) \lim_{n \rightarrow \infty} \frac{3n}{4n+5}$$

$$\begin{aligned} i) \lim_{n \rightarrow \infty} \frac{3n}{4n+5} & \quad \frac{\infty}{\infty} \\ &= \lim_{n \rightarrow \infty} \frac{\cancel{3}^1}{\cancel{4}^1 + \frac{5}{n}} \\ &= \lim_{n \rightarrow \infty} \frac{3}{4 + \frac{5}{n}} \\ &= \frac{3}{4+0} = \frac{3}{4} \end{aligned}$$

$$ii) \lim_{n \rightarrow \infty} \frac{2n+1}{n^2+n-2}$$

$$\begin{aligned} ii) \lim_{n \rightarrow \infty} \frac{2n+1}{n^2+n-2} & \quad \frac{\infty}{\infty} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{\cancel{2}^1}{\cancel{n}^2} + \frac{1}{n^2}}{\frac{\cancel{n}^2}{\cancel{n}^2} + \frac{\cancel{n}^1}{n^2} - \frac{2}{n^2}} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{2}{n} + \frac{1}{n^2}}{1 + \frac{1}{n} - \frac{2}{n^2}} = \frac{0+0}{1+0-0} = 0 \end{aligned}$$

$$iii) \lim_{n \rightarrow \infty} \frac{3n^3}{2n^2-1}$$

$$\begin{aligned} iii) \lim_{n \rightarrow \infty} \frac{3n^3}{2n^2-1} & \quad \frac{\infty}{\infty} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{\cancel{3}^3}{\cancel{n}^3}}{\frac{2\cancel{n}^2}{\cancel{n}^3} - \frac{1}{n^3}} \\ &= \lim_{n \rightarrow \infty} \frac{3}{\frac{2n}{n^2} - \frac{1}{n^3}} = \text{不存在} \end{aligned}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

g)求下列的极限:

$$i) \lim_{n \rightarrow \infty} \frac{3n}{4n+5}$$

$$ii) \lim_{n \rightarrow \infty} \frac{2n+1}{n^2+n-2}$$

$$iii) \lim_{n \rightarrow \infty} \frac{3n^3}{2n^2-1}$$

$\frac{\infty}{\infty}$ 型 总结: $\lim_{n \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{f(x)}{g(x)}$, $f(x), g(x)$ 同次方的领导系数之比

例: $\lim_{n \rightarrow \infty} \frac{3n}{4n+5} = \frac{3}{4}$

$$\lim_{n \rightarrow \infty} \frac{2n+1}{n^2+n-2} = \lim_{n \rightarrow \infty} \frac{0n^2+2n+1}{n^2+n-2} = \frac{0}{1} = 0$$

$$\lim_{n \rightarrow \infty} \frac{3n^3}{2n^2-1} = \lim_{n \rightarrow \infty} \frac{3n^3}{0n^3+2n^2-1} = \text{不存在}, (\because \text{分母} = 0)$$

随堂练习

今天高数老师有事请假了，就请食堂阿姨来给我们上一课，上了半学期，第一次发现我听懂了...

填空题：

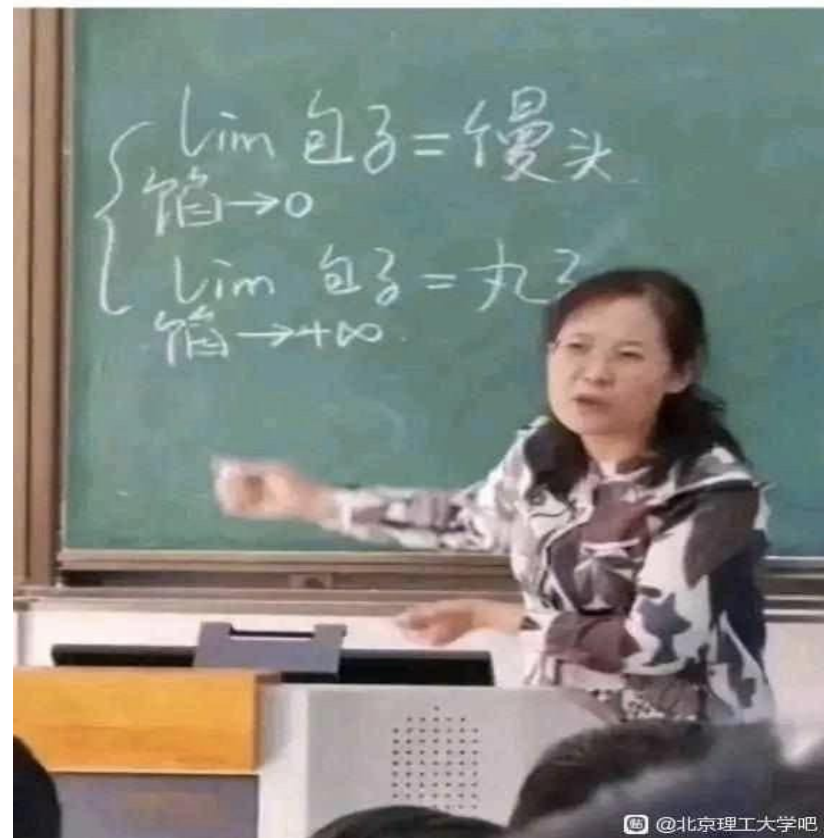
$$1) \lim_{n \rightarrow \infty} \frac{20n^{20} + 8n + 27}{4n^{20}} = ? \quad \frac{\infty}{\infty} \quad \lim_{n \rightarrow \infty} \frac{20 + \frac{8}{n^{19}} + \frac{27}{n^{20}}}{4} = \frac{20 + 0 + 0}{4} = \frac{20}{4} = 5$$

$$2) \lim_{n \rightarrow \infty} \frac{n + 27}{4n^2 + 6} = ? \quad \frac{\infty}{\infty} \quad \lim_{n \rightarrow \infty} \frac{\frac{1}{n} + \frac{27}{n^2}}{4 + \frac{6}{n^2}} = \frac{0}{4} = 0$$

$$3) \lim_{n \rightarrow \infty} \frac{n^3}{7n + 6} = ? \quad \frac{\infty}{\infty} \quad \lim_{n \rightarrow \infty} \frac{1}{\frac{7}{n^2} + \frac{6}{n^3}} = \text{不存在}$$

$$4) \lim_{n \rightarrow \infty} \frac{(n)(2n-1)(3n-1)}{(2n)(2n+1)(2n+2)} = ? \quad \frac{\infty}{\infty} \quad \lim_{n \rightarrow \infty} \frac{(1)\left(2 - \frac{1}{n}\right)\left(3 - \frac{1}{n}\right)}{(2)\left(2 + \frac{1}{n}\right)\left(2 + \frac{2}{n}\right)} = \frac{1(2)(3)}{(2)(2)(2)} = \frac{3}{4}$$

$$5) \lim_{n \rightarrow \infty} \frac{(2n+1)^5}{(n+2)^3(4n+1)^2} = ?$$



$\lim_{\text{珍珠} \rightarrow 0} \text{珍珠奶} =$

$\lim_{\text{milo} \rightarrow 0} \text{milo ice} =$

复习: $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$

h)求下列的极限:

$$i) \lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{4n^2}$$

$$i) \lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{4n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n k}{4n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{(n)(n+1)}{4n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + n}{8n^2} \quad \frac{\infty}{\infty}$$

常犯错误:

$$\begin{aligned} & i) \lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{4n^2} \\ & = \lim_{n \rightarrow \infty} \frac{\frac{1}{n^2} + \frac{2}{n^2} + \frac{3}{n^2} + \dots + \frac{1}{n}}{4} \\ & = \frac{0+0+0+\dots+0}{4} \\ & = 0 \end{aligned}$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{n^2} + \cancel{n}}{\cancel{n^2} + \cancel{n^2}}$$

$$= \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{8}$$

$$= \frac{1+0}{8} = \frac{1}{8}$$

$$ii) \lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3 + 2}$$

$$ii) \lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3 + 2}$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n k^2}{n^3 + 2}$$

$$= \lim_{n \rightarrow \infty} \frac{(n)(n+1)(2n+1)}{6n^3 + 2}$$

$$= \lim_{n \rightarrow \infty} \frac{(n)(n+1)(2n+1)}{6(n^3 + 2)} \quad \frac{\infty}{\infty}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(\frac{\cancel{n}}{\cancel{n}}\right)\left(\frac{\cancel{n}}{\cancel{n}} + \frac{1}{n}\right)\left(\frac{2\cancel{n}}{\cancel{n}} + \frac{1}{n}\right)}{6\left(\frac{\cancel{n^3}}{\cancel{n^3}} + \frac{2}{n^3}\right)}$$

$$= \lim_{n \rightarrow \infty} \frac{(1)\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right)}{6\left(1 + \frac{2}{n^3}\right)}$$

$$= \frac{(1)(1+0)(2+0)}{6(1+0)} = \frac{1}{3}$$

练习21.1 -1(a,c,e,g),4,

外加 $\lim_{n \rightarrow \infty} \frac{1+3^n}{2^n + 4^n}$

$$i) \lim_{n \rightarrow \infty} \frac{1+3+5+\dots+(2n-1)}{1+4+7+\dots+(3n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{1+3+5+\dots+(2n-1)}{1+4+7+\dots+(3n+1)}$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n (2k-1)}{1 + \sum_{k=1}^n (3k+1)}$$

$$= \lim_{n \rightarrow \infty} \frac{2 \sum_{k=1}^n k - \sum_{k=1}^n 1}{1 + 3 \sum_{k=1}^n k + \sum_{k=1}^n 1}$$

$$\text{复习: } \sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$$

$$= \lim_{n \rightarrow \infty} \frac{2 \left(\frac{n(n+1)}{2} \right) - n}{1 + 3 \left(\frac{n(n+1)}{2} \right) + n}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + n - n}{2 + 3n^2 + 3n + 2n}$$

$$= \lim_{n \rightarrow \infty} \frac{2n^2}{3n^2 + 5n + 2}$$

复习: C, D 为常数

$$\sum_{k=1}^n C = nC$$

$$\sum_{k=1}^n Ck = C \sum_{k=1}^n k$$

$$\sum_{k=1}^n (Ck \pm D) = C \sum_{k=1}^n k \pm \sum_{k=1}^n D = C \sum_{k=1}^n k \pm nD$$

$$= \lim_{n \rightarrow \infty} \frac{2}{3 + \frac{5}{n} + \frac{2}{n^2}}$$

$$= \frac{2}{3+0}$$

$$= \frac{2}{3}$$

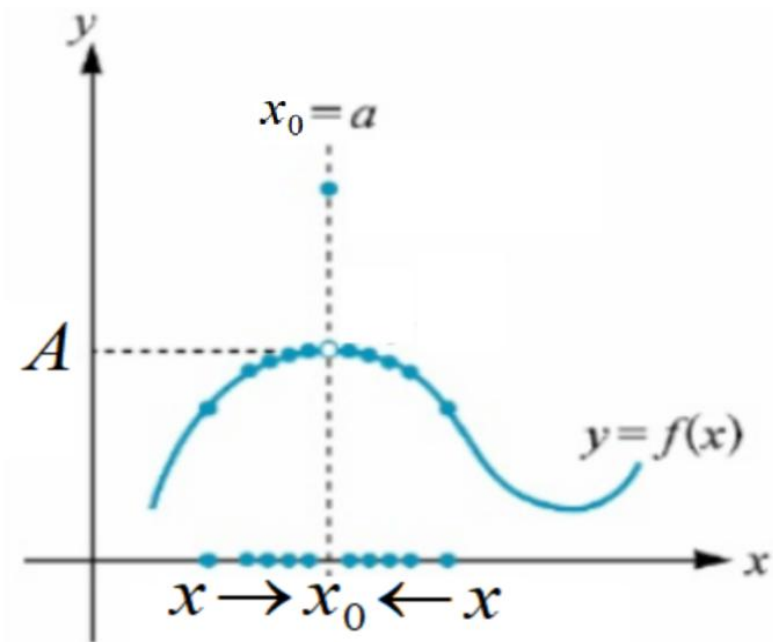
思考: $\lim_{n \rightarrow \infty} \left(3n - \frac{3n^2 + 1}{n+1} \right)$

一、 $\lim_{n \rightarrow \infty} \left(3n - \frac{3n^2 + 1}{n+1} \right) = \lim_{n \rightarrow \infty} (3n) - \lim_{n \rightarrow \infty} \frac{3n^2 + 1}{n+1} = \infty - \infty = 0$

二、 $\lim_{n \rightarrow \infty} \left(3n - \frac{3n^2 + 1}{n+1} \right) = \lim_{n \rightarrow \infty} \frac{3n^2 + 3n - (3n^2 + 1)}{n+1} = \lim_{n \rightarrow \infty} \frac{3n-1}{n+1} = \lim_{n \rightarrow \infty} \frac{3 - \frac{1}{n}}{1 + \frac{1}{n}} = \frac{3}{1} = 3$

判断以上二种做法，何者正确？并指出错误地方。

Chp21.2 函数的极限



当 $x \rightarrow x_0$, 其中 $f(x) \rightarrow A$, 记作 $\lim_{x \rightarrow x_0} f(x) = A$

$\lim_{x \rightarrow x_0} f(x)$ 极限存在的条件:

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = A$$

左极限 右极限

当 x “左、右” 无限趋近 x_0 (即 $x \rightarrow x_0^-$, $x \rightarrow x_0^+$, 但 $x \neq x_0$);

其中函数 $f(x)$ 的值 “左、右” 无限趋近 A (即 $f(x_0^-) \rightarrow A$, $f(x_0^+) \rightarrow A$),

则称 $f(x)$ 的极限值是 A , 记作: $\lim_{x \rightarrow x_0} f(x) = A$

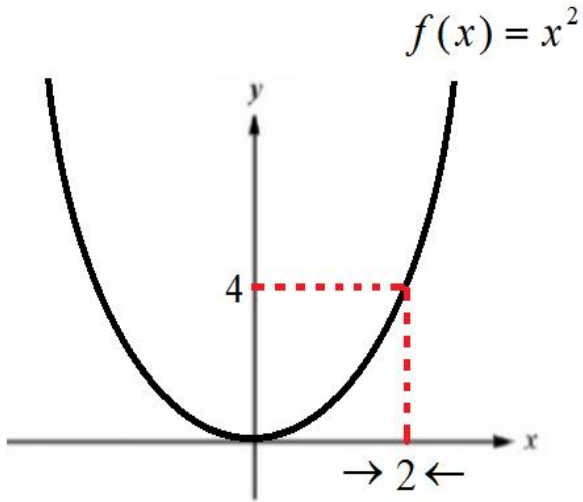
a)研究当 $x \rightarrow 2$ 时,函数 $f(x) = x^2$ 的变化趋势, 并求 $\lim_{x \rightarrow 2} x^2$ 。

x	1.9	1.99	1.999	1.9999	1.99999	...	趋近 2
$f(x) = x^2$	3.61	3.96	3.996	3.9996	3.99996	...	趋近 4

左极限, $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} x^2 = 4$

x	2.1	2.01	2.001	2.0001	2.00001	...	趋近 2
$f(x) = x^2$	4.41	4.04	4.004	4.0004	4.00004	...	趋近 4

右极限, $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} x^2 = 4$



$\therefore \lim_{x \rightarrow 2^+} x^2 = \lim_{x \rightarrow 2^-} x^2$

$\therefore \lim_{x \rightarrow 2} x^2 = 4$

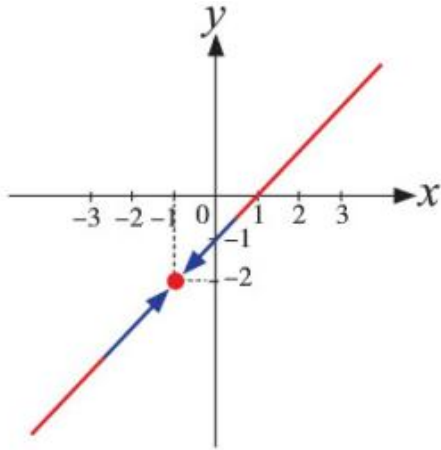
b)研究当 $x \rightarrow -1$ 时,函数 $f(x) = \frac{x^2 - 1}{x + 1}$ 的变化趋势, 并求 $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1}$ 。

x	-1.1	-1.01	-1.001	-1.0001	-1.00001	...
$f(x)$	-2.1	-2.01	-2.001	-2.0001	-2.00001	...

左极限, $\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} \frac{x^2 - 1}{x + 1} = -2$

x	-0.9	-0.99	-0.999	-0.9999	-0.99999	...
$f(x)$	-1.9	-1.99	-1.999	-1.9999	-1.99999	...

右极限, $\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} \frac{x^2 - 1}{x + 1} = -2$



$\therefore \lim_{x \rightarrow -1^+} \frac{x^2 - 1}{x + 1} = \lim_{x \rightarrow -1^-} \frac{x^2 - 1}{x + 1}$

$\therefore \lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1} = -2$

Chp21.3 函数极限的运算法则

$$\lim_{x \rightarrow x_0} k = k$$

$$\lim_{x \rightarrow x_0} x = x_0$$

若 $\lim_{x \rightarrow x_0} f(x), \lim_{x \rightarrow x_0} g(x)$ 存在, k 为常数

$$i) \lim_{x \rightarrow x_0} [kf(x)] = k \lim_{x \rightarrow x_0} f(x)$$

$$ii) \lim_{x \rightarrow x_0} [f(x) \pm g(x)] = \lim_{x \rightarrow x_0} f(x) \pm \lim_{x \rightarrow x_0} g(x)$$

$$iii) \lim_{x \rightarrow x_0} [f(x) \cdot g(x)] = \lim_{x \rightarrow x_0} f(x) \cdot \lim_{x \rightarrow x_0} g(x)$$

$$iv) \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)}, \text{ 其中 } \lim_{x \rightarrow x_0} g(x) \neq 0$$

回顾

研究当 $x \rightarrow 2$ 时, 函数 $f(x) = x^2$ 的变化趋势, 并求 $\lim_{x \rightarrow 2} x^2$ 。

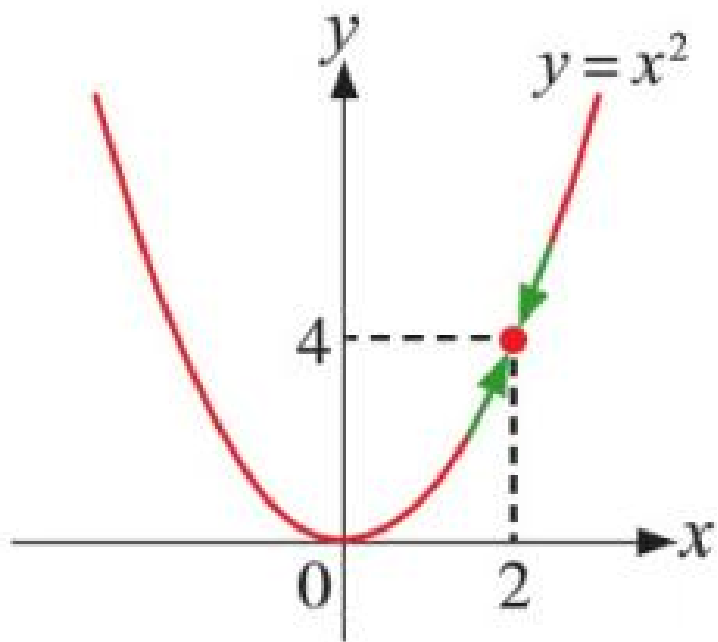
解一:

x	1.9	1.99	1.999	1.9999	1.99999	...	趋近 2
$f(x) = x^2$	3.61	3.96	3.996	3.9996	3.99996	...	趋近 4

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} x^2 = 4$$

x	2.1	2.01	2.001	2.0001	2.00001	...	趋近 2
$f(x) = x^2$	4.41	4.04	4.004	4.0004	4.00004	...	趋近 4

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} x^2 = 4$$



$$\therefore \lim_{x \rightarrow 2^+} x^2 = \lim_{x \rightarrow 2^-} x^2$$

$$\therefore \lim_{x \rightarrow 2} x^2 = 4$$

解二: $x \rightarrow 2$ (无限靠近 2 的数, 即 1.99999999 999)
则 $x^2 \rightarrow 4$ (即 $1.99999999\ 999^2 = 3.99999999\ \dots$)
即 $\lim_{x \rightarrow 2} x^2 = 4$ 。 **直接代 $x=2$ 即可**

$$a) \lim_{x \rightarrow 1} (3x + 7) = 10$$

$$b) \lim_{x \rightarrow 4} \frac{1}{x + 2} = \frac{1}{6}$$

$$c) \lim_{x \rightarrow \infty} \frac{1}{x + 2} = 0$$

$$d) \lim_{x \rightarrow 1} \frac{2x}{x - 1} = \text{不存在}$$

$$e) \lim_{x \rightarrow 1} \frac{2x - 2}{x + 1} = 0$$

练习21.3(1) -1, 4, 5, 9, 13

总结：

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)}$$

step1: 代 $x = x_0$ 入 $f(x)$ 、 $g(x)$

$$\text{step2: 若 } \left\{ \begin{array}{l} \frac{0}{\text{常数}} = 0 \\ \frac{\text{常数}}{\text{常数}} = \text{不存在} \\ \frac{0}{\text{常数}} = 0 \\ \frac{\infty}{\text{常数}} = \text{不存在} \\ \frac{\infty}{\infty} \text{型, (称“不定型”)} \\ \frac{0}{0} \text{型, (称“不定型”)} \\ \infty - \infty \text{型, (称“不定型”)} \end{array} \right.$$

若直接代值，出现 $\frac{\infty}{\infty}$ 类型

$\frac{\infty}{\infty}$ 型	法 1: 同除以分子或分母最高次方	$\left\{ \begin{array}{l} \text{若分子} = \text{分母}, = \text{比较同次方的领导系数} \\ \text{若分子} < \text{分母}, = 0 \\ \text{若分子} > \text{分母}, = \text{不存在} \end{array} \right.$
	法 2: 洛比达法则: $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{f''(x)}{g''(x)} = \dots$	

$$\begin{aligned}
 a) \lim_{x \rightarrow \infty} \frac{x^2 + 1}{2x^2 + x + 2} & \quad \frac{\infty}{\infty} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{\cancel{x^2}}{\cancel{x^2}} + \frac{1}{x^2}}{\frac{2\cancel{x^2}}{\cancel{x^2}} + \frac{\cancel{x}}{x^2} + \frac{2}{x^2}} \\
 &= \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x^2}}{2 + \frac{1}{x} + \frac{2}{x^2}} = \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 b) \lim_{x \rightarrow \infty} \frac{2x^2}{x^3 + x - 9} & \quad \frac{\infty}{\infty} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{2\cancel{x^2}}{\cancel{x^2}}}{\frac{\cancel{x^3}}{\cancel{x^3}} + \frac{\cancel{x}}{x^2} - \frac{9}{x^3}} \\
 &= \lim_{x \rightarrow \infty} \frac{2}{1 + \frac{1}{x^2} - \frac{9}{x^3}} = 0
 \end{aligned}$$

$$\begin{aligned}
 c) \lim_{x \rightarrow \infty} \frac{x^3 + 2x + 1}{x^2 + 3x - 2} & \quad \frac{\infty}{\infty} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{\cancel{x^3}}{\cancel{x^3}} + \frac{2\cancel{x^2}}{x^2} + \frac{1}{x^3}}{\frac{\cancel{x^2}}{\cancel{x^2}} + \frac{3\cancel{x}}{x^2} - \frac{2}{x^3}} \\
 &= \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x} + \frac{1}{x^3}}{\frac{1}{x} + \frac{3}{x^2} - \frac{2}{x^3}} = \text{不存在}
 \end{aligned}$$

若直接代值, 出现 $\frac{\infty}{\infty}$ 类型

$\frac{\infty}{\infty}$ 型

法 1: 同除以分子或分母最高次方

$\begin{cases} \text{若分子} = \text{分母}, = \text{比较同次方的领导系数} \\ \text{若分子} < \text{分母}, = 0 \\ \text{若分子} > \text{分母}, = \text{不存在} \end{cases}$

法 2: 洛比达法则: $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{f''(x)}{g''(x)} = \dots$

$$d) \lim_{x \rightarrow \infty} \frac{x^2 + 1}{4x} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} + \frac{1}{x^2}}{\frac{4x}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x^2}}{\frac{4}{x}}$$

= 不存在

$$e) \lim_{x \rightarrow \infty} \frac{3x^3 + 8x - 1}{x^3 + x - 9} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3x^3}{x^3} + \frac{8x}{x^3} - \frac{1}{x^3}}{\frac{x^3}{x^3} + \frac{x}{x^3} - \frac{9}{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{3 + \frac{8}{x^2} - \frac{1}{x^3}}{1 + \frac{1}{x^2} - \frac{9}{x^3}} = \frac{3 + 0 - 0}{1 + 0 - 0} = 3$$

$$f) \lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^4 + 3x} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^4} - \frac{1}{x^4}}{\frac{x^4}{x^4} + \frac{3x}{x^4}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} - \frac{1}{x^4}}{1 + \frac{3}{x^3}} = \frac{0 - 0}{1 + 0} = 0$$

若直接代值，出现 $\frac{\infty}{\infty}$ 类型

$\frac{\infty}{\infty}$ 型

法1: 同除以分子或分母最高次方

$\begin{cases} \text{若分子=分母, =比较同次方的领导系数} \\ \text{若分子<分母, =0} \\ \text{若分子>分母, =不存在} \end{cases}$

法2: 洛比达法则: $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{f''(x)}{g''(x)} = \dots$

$$g) \lim_{x \rightarrow \infty} \frac{(x+2)(2x+3)(3x+4)(4x+5)}{(1+x)(1-x)(1+x^2)}$$

$\frac{\infty}{\infty}$

分子、分母最高次方为4次，故分子、分母同除以 x^4

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \frac{\left(\frac{\cancel{x}}{\cancel{x}} + \frac{2}{x}\right) \left(\frac{2\cancel{x}}{\cancel{x}} + \frac{3}{x}\right) \left(\frac{3\cancel{x}}{\cancel{x}} + \frac{4}{x}\right) \left(\frac{4\cancel{x}}{\cancel{x}} + \frac{5}{x}\right)}{\left(\frac{1}{x} + \frac{\cancel{x}}{\cancel{x}}\right) \left(\frac{1}{x} - \frac{\cancel{x}}{\cancel{x}}\right) \left(\frac{1}{x^2} + \frac{\cancel{x^2}}{\cancel{x^2}}\right)} \\
 &= \lim_{x \rightarrow \infty} \frac{\left(1 + \frac{2}{x}\right) \left(2 + \frac{3}{x}\right) \left(3 + \frac{4}{x}\right) \left(4 + \frac{5}{x}\right)}{\left(\frac{1}{x} + 1\right) \left(\frac{1}{x} - 1\right) \left(\frac{1}{x^2} + 1\right)} = \frac{(1)(2)(3)(4)}{(1)(-1)(1)} = -24
 \end{aligned}$$

若直接代值，出现 $\frac{\infty}{\infty}$ 类型

$$h) \lim_{x \rightarrow \infty} \frac{\sqrt{16x^2 - 1} + x}{2x} \quad \frac{\infty}{\infty}$$

分子、分母最高次方为1次，故分子、分母同除以 x

$$= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{16x^2 - 1}}{x} + \frac{x}{x}}{2x}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{16x^2}{x^2} - \frac{1}{x^2}} + 1}{2x^2}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{16 - \frac{1}{x^2}} + 1}{2}$$

$$= \frac{\sqrt{16} + 1}{2}$$

$$= \frac{5}{2}$$

$\frac{\infty}{\infty}$ 型

法1: 同除以分子或分母最高次方

若分子=分母，=比较同次方的领导系数
若分子<分母，=0
若分子>分母，=不存在

法2: 洛比达法则: $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{f''(x)}{g''(x)} = \dots$



IF:

$$\lim_{x \rightarrow 8} \frac{1}{x-8} = \infty$$

THEN:

$$\lim_{x \rightarrow 5} \frac{1}{x-5} = \infty$$

练习21.3(2) -18,19,20,24,

外加 “ $\lim_{x \rightarrow \infty} \frac{ax^3 + bx^2 + 4x - 1}{5x^2 + x - 1} = 3$, 求 a, b 的值。”

随堂练习(填空题)

$$2. \lim_{x \rightarrow -3} (x^2 - 2x + 7) = 22$$

$$4. \lim_{x \rightarrow 1} x(9 - 2x^2) = 7$$

$$8. \lim_{x \rightarrow 3} \frac{x^2 - 1}{x + 1} = 2$$

$$12. \lim_{x \rightarrow 1} \frac{2x^2}{3x^2 + 4x - 1} = \frac{1}{3}$$

$$16. \lim_{x \rightarrow 0} \frac{1}{x^2} = \text{不存在}$$

$$18. \lim_{x \rightarrow 1} \frac{4x - 3}{x^2 - 5x + 4} = \text{不存在}$$

$$20. \lim_{x \rightarrow 1} \frac{2x^2 - 1}{3x^3 - 6x^2 + 5} = \frac{1}{2}$$

$$\frac{\infty}{\infty} 40. \lim_{x \rightarrow \infty} \frac{x^2 + 1}{2x^2 + 2x - 1} = \frac{1}{2}$$

$$\frac{\infty}{\infty} 42. \lim_{x \rightarrow \infty} \frac{2x^3 + 1}{3x^3 - 2} = \frac{2}{3}$$

$$\frac{\infty}{\infty} 44. \lim_{x \rightarrow \infty} \frac{x + x^2 - 6x^3}{2x - 5x^2 - 3x^3} = 2$$

$$\frac{\infty}{\infty} 46. \lim_{x \rightarrow \infty} \frac{x^4 - 2x^3 + x^2 + 3}{x^5 - x^4 + 1} = 0$$

回顾

研究当 $x \rightarrow -1$ 时, 函数 $f(x) = \frac{x^2 - 1}{x + 1}$ 的变化趋势, 并求 $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1}$ 。

解:

x	-1.1	-1.01	-1.001	-1.0001	-1.00001	...
$f(x)$	-2.1	-2.01	-2.001	-2.0001	-2.00001	...

x	-0.9	-0.99	-0.999	-0.9999	-0.99999	...
$f(x)$	-1.9	-1.99	-1.999	-1.9999	-1.99999	...

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} \frac{x^2 - 1}{x + 1} = -2$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} \frac{x^2 - 1}{x + 1} = -2$$

解二: $x \rightarrow -1$ (无限靠近 -1 的数, 即 -1.00000001)

$$\therefore \lim_{x \rightarrow -1^-} \frac{x^2 - 1}{x + 1} = \lim_{x \rightarrow -1^+} \frac{x^2 - 1}{x + 1}$$

$$\therefore \lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1} = -2$$

$$\text{则 } \frac{x^2 - 1}{x + 1} = \frac{(x + 1)(x - 1)}{x + 1} = x - 1 \rightarrow -2$$

$$(\text{即 } \frac{(-1.00000001)^2 - 1}{-1.00000001 + 1} = -2.00000001 \dots)$$

$$\text{即 } \lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1} = -2.$$

先因式分解,
再代 $x = -1$ 即可

若直接代值，出现 $\frac{0}{0}$ 类型

$\frac{0}{0}$ 型	法 1: $\begin{cases} \text{有理式, 因式分解} \\ \text{无理式, 有理化} \end{cases}$
	法 2: 洛比达法则: $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)} = \lim \frac{f''(x)}{g''(x)} = \dots$

$$a) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{(x+1)\cancel{(x-1)}}{\cancel{x-1}} \quad \begin{array}{l} \text{无 } \sqrt{}, \\ \text{因式分解} \end{array}$$

$$= \lim_{x \rightarrow 1} (x + 1)$$

$$= 1 + 1$$

$$= 2$$

$$b) \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} \times \frac{\sqrt{x} + 1}{\sqrt{x} + 1} \quad \begin{array}{l} \text{有 } \sqrt{}, \\ \text{有理化} \end{array}$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{x-1}}{(\cancel{x-1})(\sqrt{x} + 1)}$$

$$= \lim_{x \rightarrow 1} \frac{1}{\sqrt{x} + 1} = \frac{1}{1 + 1} = \frac{1}{2}$$

若直接代值，出现 $\frac{0}{0}$ 类型

$\frac{0}{0}$ 型	法 1: $\begin{cases} \text{有理式, 因式分解} \\ \text{无理式, 有理化} \end{cases}$
	法 2: 洛比达法则: $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)} = \lim \frac{f''(x)}{g''(x)} = \dots$

$$c) \lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x^2 + 2x - 3} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(x-2)}{\cancel{(x-1)}(x+3)}$$

$$= \frac{1-2}{1+3}$$

$$= -\frac{1}{4}$$

$$d) \lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x^2 + 4x + 4} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow -2} \frac{\cancel{(x+2)}(x+3)}{\cancel{(x+2)}(x+2)}$$

= 不存在

$$e) \lim_{x \rightarrow 2} (x-2) \frac{x}{x^2 - 4} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 2} \cancel{(x-2)} \frac{x}{(x+2)\cancel{(x-2)}}$$

$$= \frac{2}{2+2}$$

$$= \frac{1}{2}$$

若直接代值，出现 $\frac{0}{0}$ 类型

$\frac{0}{0}$ 型	法 1: $\begin{cases} \text{有理式, 因式分解} \\ \text{无理式, 有理化} \end{cases}$
	法 2: 洛比达法则: $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)} = \lim \frac{f''(x)}{g''(x)} = \dots$

$$\begin{aligned} f) \lim_{x \rightarrow 0} \frac{3x^3 + 4x^2 + 5x}{5x^3 + 4x^2 + 3x} & \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{\cancel{x} (3x^2 + 4x + 5)}{\cancel{x} (5x^2 + 4x + 3)} \\ &= \frac{0 + 0 + 5}{0 + 0 + 3} \\ &= \frac{5}{3} \end{aligned}$$

$$\begin{aligned} g) \lim_{x \rightarrow 1} \frac{x^4 + 3x^3 - 2x^2 - 6x + 4}{2x^3 - x^2 + 3x - 4} & \frac{0}{0} \\ &= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)} (x^3 + 4x^2 + 2x - 4)}{\cancel{(x-1)} (2x^2 + x + 4)} \\ &= \frac{1 + 4 + 2 - 4}{2 + 1 + 4} \\ &= \frac{3}{7} \end{aligned}$$

若直接代值，出现 $\frac{0}{0}$ 类型

$\frac{0}{0}$ 型	法 1: $\begin{cases} \text{有理式, 因式分解} \\ \text{无理式, 有理化} \end{cases}$
	法 2: 洛比达法则: $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)} = \lim \frac{f''(x)}{g''(x)} = \dots$

$$\begin{aligned}
 h) \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x-1} & \quad \frac{0}{0} \\
 = \lim_{x \rightarrow 1} \left(\frac{\sqrt{x+3} - 2}{x-1} \times \frac{\sqrt{x+3} + 2}{\sqrt{x+3} + 2} \right) \\
 = \lim_{x \rightarrow 1} \frac{(\sqrt{x+3})^2 - 2^2}{(x-1)(\sqrt{x+3} + 2)} \\
 = \lim_{x \rightarrow 1} \frac{\cancel{x-1}}{(\cancel{x-1})(\sqrt{x+3} + 2)} \\
 = \frac{1}{\sqrt{1+3} + 2} = \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 i) \lim_{x \rightarrow -2} \frac{x+2}{x + \sqrt{x+6}} & \quad \frac{0}{0} \\
 = \lim_{x \rightarrow -2} \left(\frac{x+2}{x + \sqrt{x+6}} \times \frac{x - \sqrt{x+6}}{x - \sqrt{x+6}} \right) \\
 = \lim_{x \rightarrow -2} \frac{(x+2)(x - \sqrt{x+6})}{x^2 - x - 6} \\
 = \lim_{x \rightarrow -2} \frac{(\cancel{x+2})(x - \sqrt{x+6})}{(\cancel{x+2})(x-3)} = \frac{-2 - \sqrt{-2+6}}{-2-3} = \frac{4}{5}
 \end{aligned}$$

若直接代值，出现 $\frac{0}{0}$ 类型

$\frac{0}{0}$ 型	法 1: $\begin{cases} \text{有理式, 因式分解} \\ \text{无理式, 有理化} \end{cases}$
	法 2: 洛比达法则: $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)} = \lim \frac{f''(x)}{g''(x)} = \dots$

$$j) \lim_{x \rightarrow -1} \frac{\sqrt{x+4} - \sqrt{3}}{\sqrt{x+2} - 1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow -1} \left(\frac{\sqrt{x+4} - \sqrt{3}}{\sqrt{x+2} - 1} \times \frac{\sqrt{x+4} + \sqrt{3}}{\sqrt{x+4} + \sqrt{3}} \times \frac{\sqrt{x+2} + 1}{\sqrt{x+2} + 1} \right)$$

$$= \lim_{x \rightarrow -1} \left(\frac{(\sqrt{x+4})^2 - (\sqrt{3})^2}{(\sqrt{x+2})^2 - 1^2} \times \frac{\sqrt{x+2} + 1}{\sqrt{x+4} + \sqrt{3}} \right)$$

$$= \lim_{x \rightarrow -1} \left(\frac{\cancel{x+1}}{\cancel{x+1}} \times \frac{\sqrt{x+2} + 1}{\sqrt{x+4} + \sqrt{3}} \right) = \frac{\sqrt{-1+2} + 1}{\sqrt{-1+4} + \sqrt{3}} = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

若直接代值，出现 $\frac{0}{0}$ 类型

$$k) \lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{x - 1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \left(\frac{\sqrt[3]{x} - 1}{x - 1} \times \frac{\sqrt[3]{x^2} + \sqrt[3]{x} + 1}{\sqrt[3]{x^2} + \sqrt[3]{x} + 1} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(\sqrt[3]{x})^3 - 1^3}{(x - 1)(\sqrt[3]{x^2} + \sqrt[3]{x} + 1)}$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{x-1}}{(\cancel{x-1})(\sqrt[3]{x^2} + \sqrt[3]{x} + 1)} = \frac{1}{1+1+1} = \frac{1}{3}$$

$\frac{0}{0}$ 型	法 1: $\begin{cases} \text{有理式, 因式分解} \\ \text{无理式, 有理化} \end{cases}$
	$hint: \alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2) = \dots$
	即 $(\sqrt[3]{x})^3 - 1^3 = (\sqrt[3]{x} - 1)(\sqrt[3]{x^2} + \sqrt[3]{x} + 1)$

练习21.3(3) -2,3,8,12,14,16,

外加 “ $\lim_{x \rightarrow 3} \frac{x^2 - x - a}{x - 3} = b$, 求 a, b 的值。”

计算机求极限

$$1) \lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 1}{4x^2 + 1}$$

A.0 B. $\frac{3}{4}$ C. $1\frac{1}{2}$ D. ∞

$$2) \lim_{x \rightarrow \infty} \frac{x^3}{7x + 6}$$

A. $\frac{2}{5}$ B.1 C.0 D. ∞

$$3) \lim_{x \rightarrow \infty} \frac{2x^2 - x - 1}{x^2 - 1}$$

A.2 B. $\frac{3}{2}$ C. $\frac{1}{2}$ D. ∞

$$4) \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} = ?$$

A.0 B.1 C.2 D.3

$$5) \lim_{x \rightarrow 2} \frac{3x^2 - 4x - 4}{x^2 - 4}$$

A.3 B.2 C.1 D. $\frac{3}{4}$

$$6) \lim_{x \rightarrow \infty} \frac{x^2 - 6}{(x^2 - 1)(x + 2)}$$

A.0 B.1 C.3 D. ∞

计算机求极限

$$7) \lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x} - x)$$

A. $\frac{1}{2}$ B. 1 C. 0 D. ∞

$$8) \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{\sqrt{x} - 1}$$

A. $\frac{1}{2}$ B. 1 C. 0 D. ∞

$$9) \lim_{n \rightarrow \infty} \frac{1 + 2 + 3 + \dots + n}{n^2}$$

A. 1 B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{1}{4}$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$$

$$9) \lim_{n \rightarrow \infty} \frac{1 + 2 + 3 + \dots + n}{n^2} = \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n k}{n^2} = \lim_{n \rightarrow \infty} \frac{\frac{n(n+1)}{2}}{n^2}$$

若直接代值，出现 $\infty - \infty$ 类型

$$a) \lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{2}{x^2-1} \right) \quad \infty - \infty$$

$$= \lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{2}{(x+1)(x-1)} \right)$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{x+1} - 2}{(\cancel{x-1})(x+1)} \quad \begin{array}{l} \text{无}\sqrt{}, \\ \text{通分} \end{array}$$

$$= \frac{1}{1+1}$$

$$= \frac{1}{2}$$

$$b) \lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x} - x) \quad \infty - \infty$$

$$= \lim_{x \rightarrow \infty} \left[(\sqrt{x^2 + 2x} - x) \times \frac{\sqrt{x^2 + 2x} + x}{\sqrt{x^2 + 2x} + x} \right] \quad \begin{array}{l} \text{有}\sqrt{}, \\ \text{有理化} \end{array}$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 2x})^2 - x^2}{\sqrt{x^2 + 2x} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{2x}{\sqrt{x^2 + 2x} + x} = \lim_{x \rightarrow \infty} \frac{\frac{2\cancel{x}}{\cancel{x}}}{\sqrt{\frac{\cancel{x}^2}{\cancel{x}^2} + \frac{2\cancel{x}}{\cancel{x}^2}} + \frac{\cancel{x}}{\cancel{x}}} = \frac{2}{1+1} = 1$$

$\infty - \infty$ 型	法1: $\begin{cases} \text{有理式, 通分} \\ \text{无理式, 有理化} \end{cases}$
	法2: 把 $f(x) - g(x)$ 转成 $\frac{\frac{1}{g(x)} - \frac{1}{f(x)}}{\frac{1}{f(x)g(x)}}$, 变为 $\frac{0}{0}$ 型

若直接代值, 出现 $\infty - \infty$ 类型

$$c) \lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{4}{x^2-4} \right) \quad \infty - \infty$$

$$= \lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{4}{(x+2)(x-2)} \right)$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{x+2}-4}{(x+2)\cancel{(x-2)}}$$

$$= \frac{1}{2+2}$$

$$= \frac{1}{4}$$

$$d) \lim_{x \rightarrow \infty} \left(\frac{x^3}{2x^2+1} - \frac{x^2}{2x+3} \right) \quad \infty - \infty$$

$$= \lim_{x \rightarrow \infty} \frac{2x^4 + 3x^3 - 2x^4 - x^2}{(2x^2+1)(2x+3)} = \lim_{x \rightarrow \infty} \frac{3x^3 - x^2}{(2x^2+1)(2x+3)}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3\cancel{x^3}}{\cancel{x^3}} - \frac{\cancel{x^2}}{x^{\cancel{3}}}}{\left(\frac{2\cancel{x^2}}{\cancel{x^2}} + \frac{1}{x^2} \right) \left(\frac{2\cancel{x}}{\cancel{x}} + \frac{3}{x} \right)} = \frac{3}{(2)(2)} = \frac{3}{4}$$

$\infty - \infty$ 型	法1: $\begin{cases} \text{有理式, 通分} \\ \text{无理式, 有理化} \end{cases}$
	法2: 把 $f(x) - g(x)$ 转成 $\frac{\frac{1}{g(x)} - \frac{1}{f(x)}}{\frac{1}{f(x)g(x)}}$, 变为 $\frac{0}{0}$ 型

若直接代值，出现 $\infty - \infty$ 类型

$\infty - \infty$ 型	法1: $\begin{cases} \text{有理式, 通分} \\ \text{无理式, 有理化} \end{cases}$
	法2: 把 $f(x) - g(x)$ 转成 $\frac{\frac{1}{g(x)} - \frac{1}{f(x)}}{\frac{1}{f(x)g(x)}}$, 变为 $\frac{0}{0}$ 型

$$\begin{aligned}
 e) & \lim_{x \rightarrow \infty} (\sqrt{x+1} - \sqrt{x})(\sqrt{x+1}) \quad \infty - \infty \\
 &= \lim_{x \rightarrow \infty} \left[(\sqrt{x+1} - \sqrt{x}) \times \frac{\sqrt{x+1} + \sqrt{x}}{\sqrt{x+1} + \sqrt{x}} \right] \times (\sqrt{x+1}) \\
 &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x+1})^2 - (\sqrt{x})^2}{\sqrt{x+1} + \sqrt{x}} \times (\sqrt{x+1}) \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt{x+1}}{\sqrt{x+1} + \sqrt{x}} \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt{\cancel{x} + \frac{1}{\cancel{x}}}}{\sqrt{\cancel{x} + \frac{1}{\cancel{x}}} + \sqrt{\cancel{x}}} = \frac{\sqrt{1+0}}{\sqrt{1+0} + \sqrt{1}} = \frac{1}{2}
 \end{aligned}$$

若直接代值，出现 $\infty - \infty$ 类型

$$\begin{aligned}
 f) \lim_{x \rightarrow \infty} \frac{6}{\sqrt{x}(\sqrt{x+3} - \sqrt{x})} & \quad \infty - \infty \\
 &= \lim_{x \rightarrow \infty} \left[\frac{6}{\sqrt{x}(\sqrt{x+3} - \sqrt{x})} \times \frac{\sqrt{x+3} + \sqrt{x}}{\sqrt{x+3} + \sqrt{x}} \right] \\
 &= \lim_{x \rightarrow \infty} \frac{6(\sqrt{x+3} + \sqrt{x})}{\sqrt{x}(\cancel{x} + 3 - \cancel{x})} \\
 &= \lim_{x \rightarrow \infty} \frac{6 \left(\sqrt{\frac{\cancel{x}}{\cancel{x}} + \frac{3}{x}} + \sqrt{\frac{\cancel{x}}{\cancel{x}}} \right)}{3\sqrt{\frac{\cancel{x}}{\cancel{x}}}} = \frac{6(1+1)}{3} = 4
 \end{aligned}$$

$\infty - \infty$ 型	法1: $\begin{cases} \text{有理式, 通分} \\ \text{无理式, 有理化} \end{cases}$
	法2: 把 $f(x) - g(x)$ 转成 $\frac{\frac{1}{g(x)} - \frac{1}{f(x)}}{\frac{1}{f(x)g(x)}}$, 变为 $\frac{0}{0}$ 型

练习21.3(4) -21,22,23

$\infty - \infty$ 类型

无 $\sqrt{\quad}$ ，通分

$\frac{\infty}{\infty}$ 类型

解法 1：
同除以分子、分母最高次方

- 若分子 = 分母，领导系数
- 若分子 < 分母，0
- 若分子 > 分母，不存在

解法 2：
罗必达法则 *L'Hopital's Rule*
即 *LH's Rule*

$\frac{0}{0}$ 类型

解法 1

- 有理式，因式分解
- 无理式，有理化

解法 2：
罗必达法则 *L'Hopital's Rule*
即 *LH's Rule*

有 $\sqrt{\quad}$ ，有理化

$\frac{\infty}{\infty}$ 类型

解法 1：
同除以分子、分母最高次方

- 若分子 = 分母，领导系数
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解法 2：
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$\frac{0}{0}$ 类型

解法 1

- 有理式，因式分解
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解法 2：
罗必达法则 *L'Hopital's Rule*
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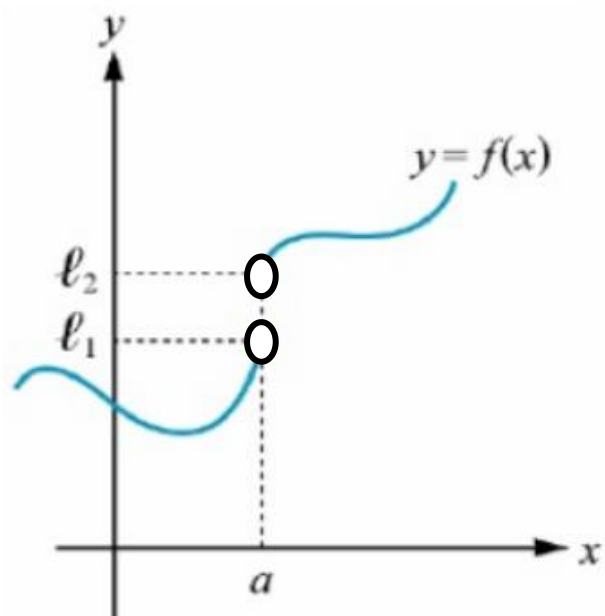
极限不存在的例子

分段函数：例子a,b

$$f(x) = \begin{cases} 1-x & , x < 1 \\ x & , x \geq 1 \end{cases}$$

绝对值函数：例子c

$$f(x) = \frac{|x|}{x}$$



$$\text{左极限: } \lim_{x \rightarrow a^-} f(x) = l_1$$

$$\text{右极限: } \lim_{x \rightarrow a^+} f(x) = l_2$$

$$\therefore \lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$$

$$\therefore \lim_{x \rightarrow a} f(x) \text{ 不存在}$$

左极限 $\neq$ 右极限

$\lim_{x \rightarrow x_0} f(x)$ 极限存在的条件:

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = A$$

$\lim_{x \rightarrow x_0} f(x)$ 极限存在的条件:

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = A$$

a) 若 $f(x) = \begin{cases} 2, & x \geq 1 \\ -2, & x < 1 \end{cases}$, 求 $\lim_{x \rightarrow 1} f(x)$ 。

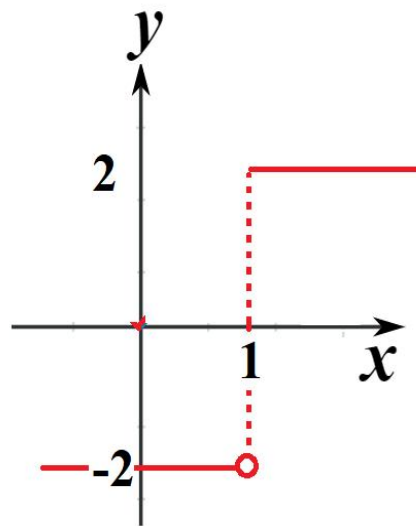
解:

左极限: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (-2) = -2$

右极限: $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 2 = 2$

$$\therefore \lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$$

$$\therefore \lim_{x \rightarrow 1} f(x) \text{ 不存在}$$



b) 若 $f(x) = \begin{cases} 10 - x, & x > 2 \\ 6, & x = 2 \\ x^2, & x < 2 \end{cases}$, 求 $\lim_{x \rightarrow 2} f(x)$ 。

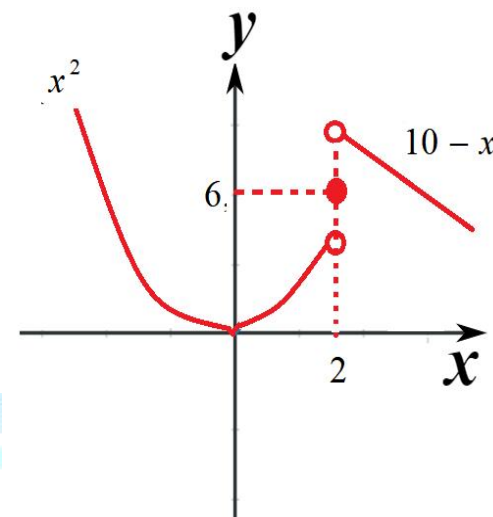
解:

左极限: $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} x^2 = 4$

右极限: $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (10 - x) = 8$

$$\therefore \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

$$\therefore \lim_{x \rightarrow 2} f(x) \text{ 不存在}$$



复习:

$$|x| = \begin{cases} x, & \text{当 } x \geq 0 \\ -x, & \text{当 } x < 0 \end{cases}$$

c) 求 $\lim_{x \rightarrow 0} \frac{|x|}{x}$ 。

解:

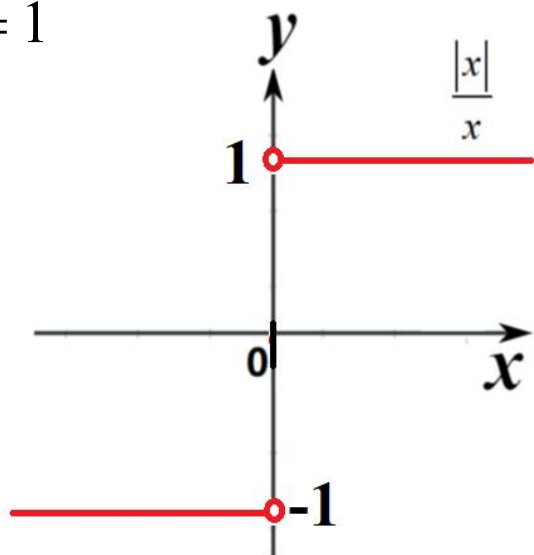
问 $|x-1| = \begin{cases} x-1, & \text{当 } x \geq ______ \\ -(x-1), & \text{当 } x < ______ \end{cases}$

左极限: $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$

右极限: $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$

$$\because \lim_{x \rightarrow 0^-} \frac{|x|}{x} \neq \lim_{x \rightarrow 0^+} \frac{|x|}{x}$$

$$\therefore \lim_{x \rightarrow 0} \frac{|x|}{x} \text{ 不存在}$$

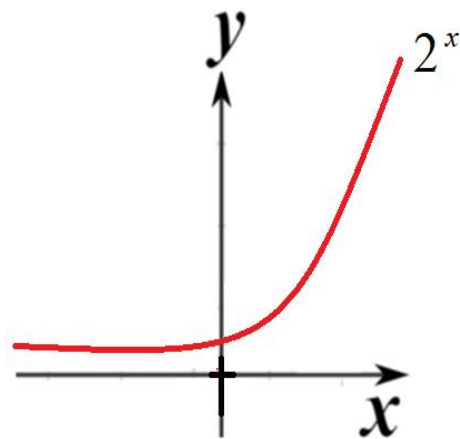


当 x 趋近正、负无穷大

d) 求 $\lim_{x \rightarrow \infty} 2^x$ 、 $\lim_{x \rightarrow -\infty} 2^x$ 。

解: $\lim_{x \rightarrow \infty} 2^x = \infty$ (不存在)

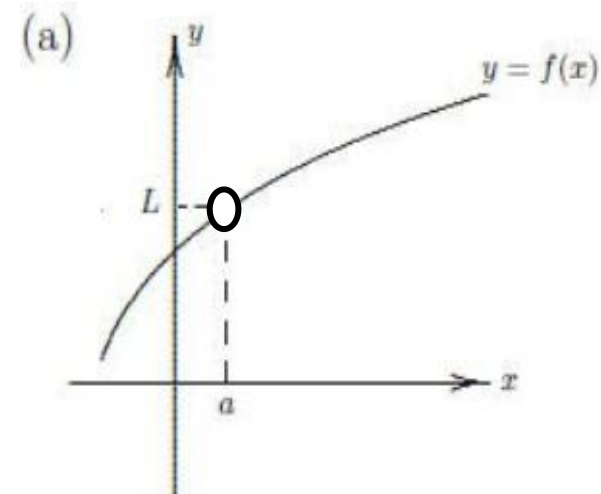
$$\lim_{x \rightarrow -\infty} 2^x = 0$$



e)求下列各小题的 $\lim_{x \rightarrow a} f(x)$ 。

$\lim_{x \rightarrow x_0} f(x)$ 极限存在的条件:

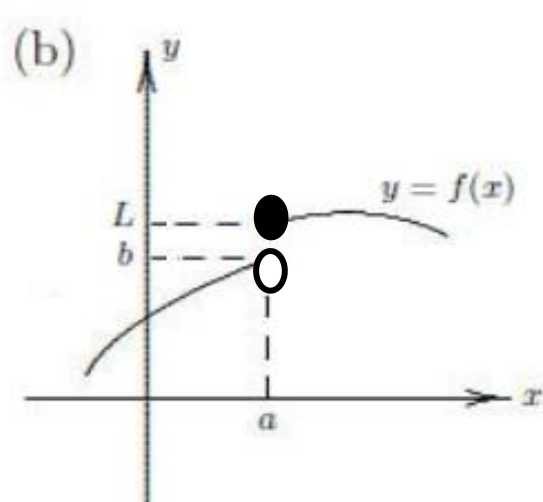
$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = A$$



$\therefore$ 左极限: $\lim_{x \rightarrow a^-} f(x) = L$

右极限: $\lim_{x \rightarrow a^+} f(x) = L$

$\therefore \lim_{x \rightarrow a} f(x) = L$

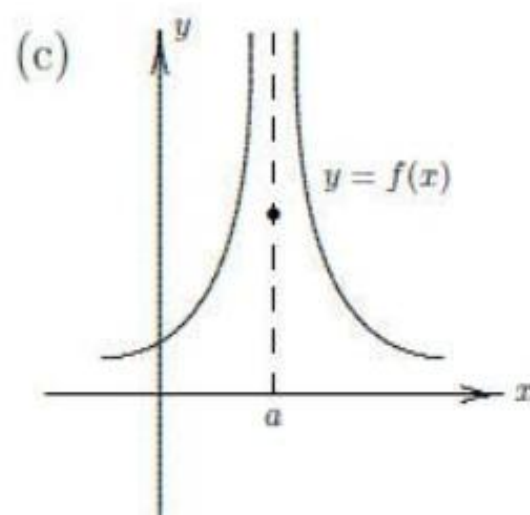


$\therefore$ 左极限: $\lim_{x \rightarrow a^-} f(x) = b$

右极限: $\lim_{x \rightarrow a^+} f(x) = L$

$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$

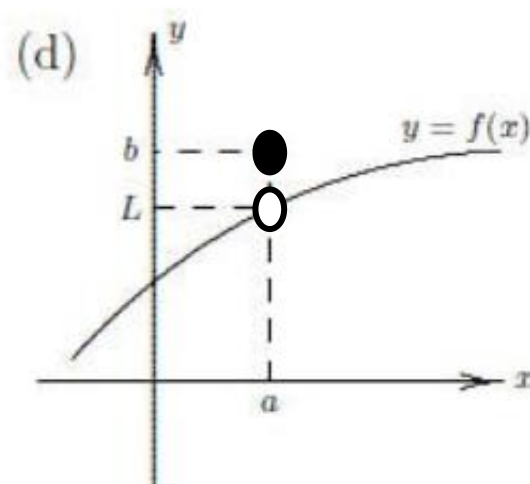
$\therefore \lim_{x \rightarrow a} f(x)$ 不存在



$\therefore$ 左极限: $\lim_{x \rightarrow a^-} f(x) = \infty$

右极限: $\lim_{x \rightarrow a^+} f(x) = \infty$

$\therefore \lim_{x \rightarrow a} f(x)$ 不存在



$\therefore$ 左极限: $\lim_{x \rightarrow a^-} f(x) = L$

右极限: $\lim_{x \rightarrow a^+} f(x) = L$

$\therefore \lim_{x \rightarrow a} f(x) = L$

习题12d-4a,c,5a,b

随堂练习

1. 若 $f(x) = \begin{cases} \frac{x^2 - x - 6}{x - 3} & , x < 3 \\ 3x & , x = 3 \\ 9 - x & , x > 3 \end{cases}$, 求 $\lim_{x \rightarrow 3} f(x)$

左极限: $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \frac{x^2 - x - 6}{x - 3}$

$$= \lim_{x \rightarrow 3^-} \frac{(x - 3)(x + 2)}{x - 3}$$
$$= 5$$

右极限: $\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (9 - x)$

$$= 6$$

$$\because \lim_{x \rightarrow 3^-} f(x) \neq \lim_{x \rightarrow 3^+} f(x)$$

$$\therefore \lim_{x \rightarrow 3} f(x) \text{ 不存在}$$

2. 若 $f(x) = \begin{cases} \frac{x^2 + x - 6}{x - 2} & , x < 2 \\ 3x & , x = 2 \\ 7 - x & , x > 2 \end{cases}$, 求 $\lim_{x \rightarrow 2} f(x)$

左极限: $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{x^2 + x - 6}{x - 2}$

$$= \lim_{x \rightarrow 2^-} \frac{(x - 2)(x + 3)}{x - 2}$$
$$= 5$$

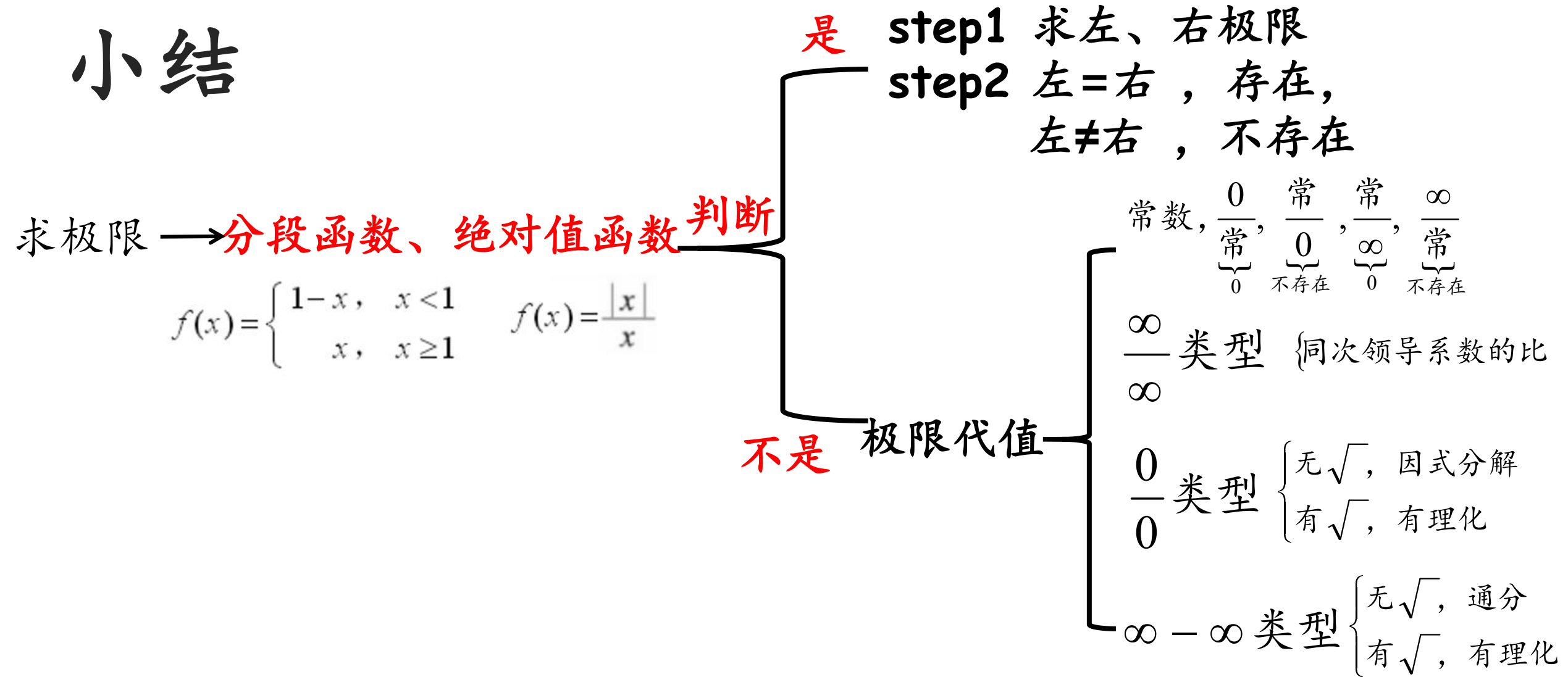
右极限: $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (7 - x)$

$$= 5$$

$$\because \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$\therefore \lim_{x \rightarrow 2} f(x) = 5$$

小结



Chp12.5 连续函数

Continuous function^{Pg. 97}

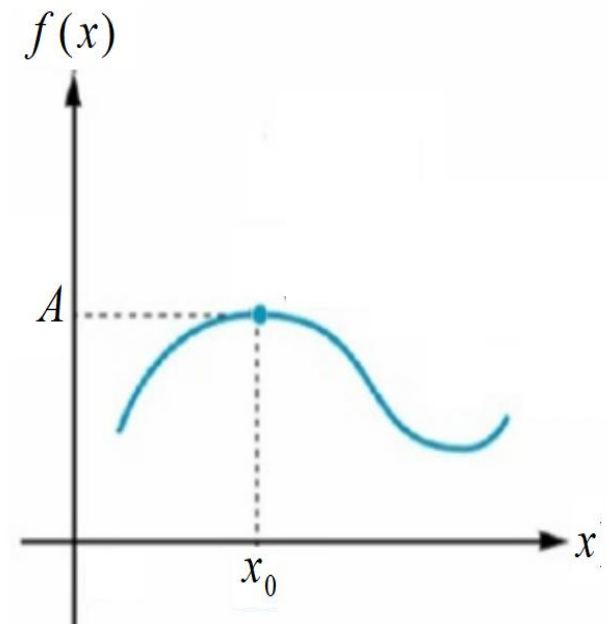
函数连续存在的条件:

(极限存在) $i) \lim_{x \rightarrow x_0} f(x) = A$

($f(x)$ 在 $x = x_0$ 处有意义) $ii) f(x_0) = A$

(极限值 = 函数值) $iii) \lim_{x \rightarrow x_0} f(x) = f(x_0)$

则 $f(x)$ 在 $x = x_0$ 处连续



$\lim_{x \rightarrow x_0} f(x)$ 极限存在的条件:

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = A$$

函数连续存在的条件:

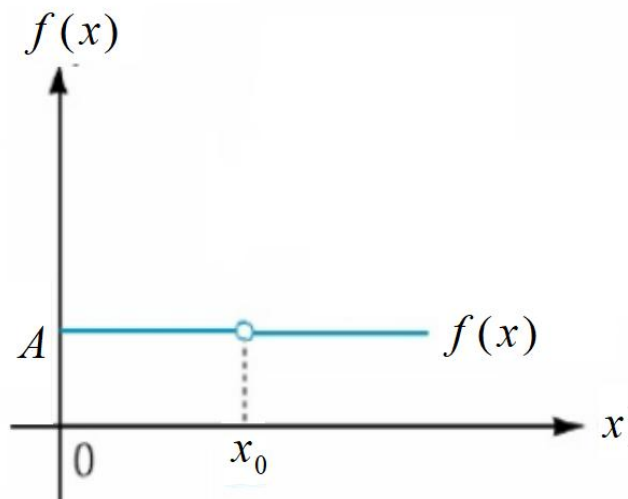
i) $\lim_{x \rightarrow x_0} f(x) = A$

ii) $f(x_0) = A$

iii) $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

则 $f(x)$ 在 $x = x_0$ 处连续

a) 判断下列个图 $f(x)$ 在 $x = x_0$ 处是否连续。

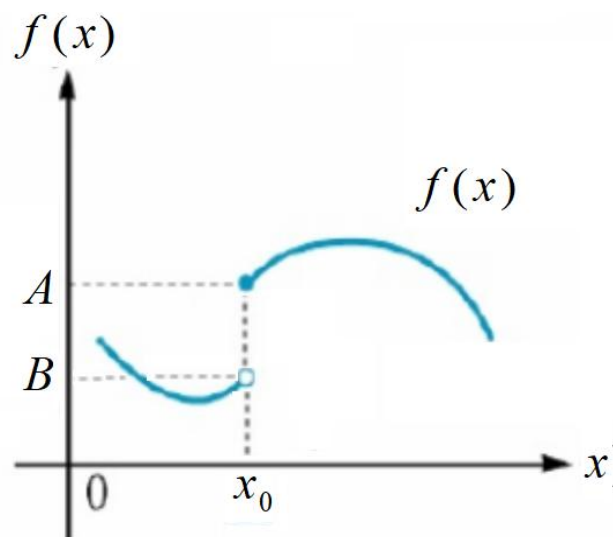


$$\because \lim_{x \rightarrow x_0^-} f(x) = A = \lim_{x \rightarrow x_0^+} f(x)$$

$$\therefore \lim_{x \rightarrow x_0} f(x) = A$$

$\because f(x_0)$ 无意义

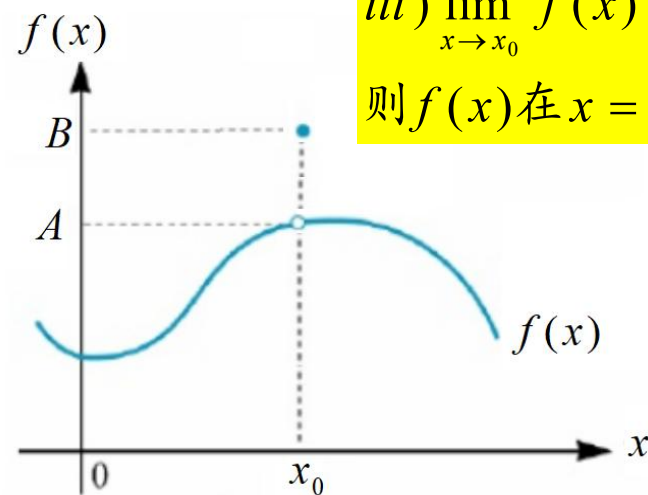
$\therefore f(x)$ 在 $x = x_0$ 处不连续



$$\because \lim_{x \rightarrow x_0^-} f(x) = B \neq A = \lim_{x \rightarrow x_0^+} f(x)$$

$\therefore \lim_{x \rightarrow x_0} f(x)$ 不存在

$\therefore f(x)$ 在 $x = x_0$ 处不连续



$$\because \lim_{x \rightarrow x_0^-} f(x) = A = \lim_{x \rightarrow x_0^+} f(x)$$

$$\therefore \lim_{x \rightarrow x_0} f(x) = A$$

$$f(x_0) = B$$

$$\because \lim_{x \rightarrow x_0} f(x) \neq f(x_0)$$

$\therefore f(x)$ 在 $x = x_0$ 处不连续

函数连续存在的条件:

i) $\lim_{x \rightarrow x_0} f(x) = A$

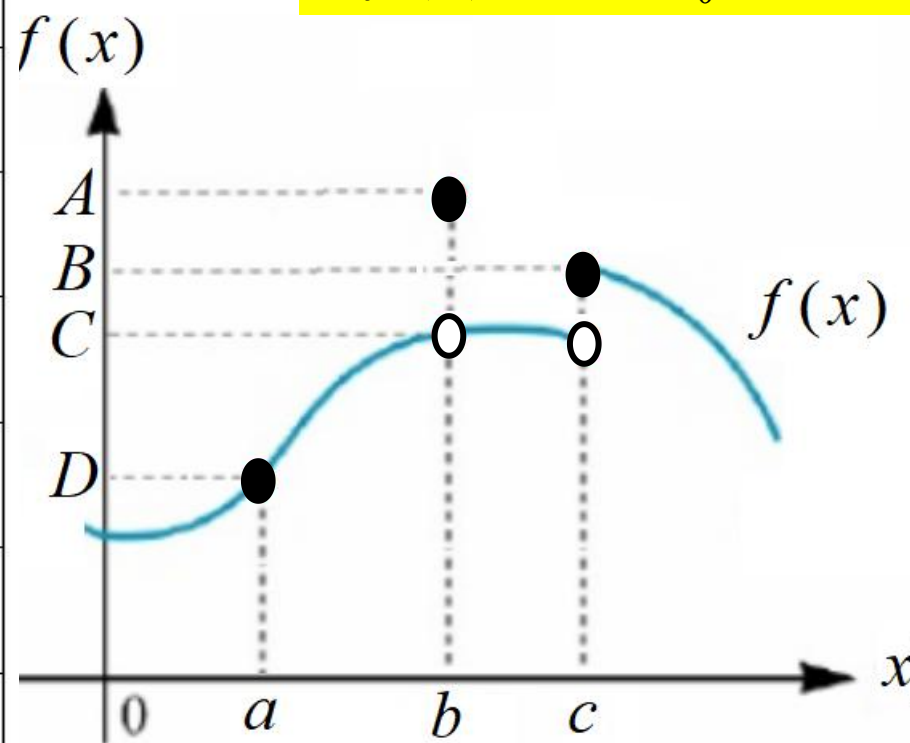
ii) $f(x_0) = A$

iii) $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

则 $f(x)$ 在 $x = x_0$ 处连续

b) 根据下图，完成下表，答案为非，必须写出原因。

	叙述	是/非	原因
1.	$\lim_{x \rightarrow a} f(x) = D$	是	
2.	$f(a) = A$	非	$f(a) = D$
3.	$f(x)$ 在 $x = a$ 连续	是	
4.	$\lim_{x \rightarrow b} f(x) = A$	非	$\lim_{x \rightarrow b} f(x) = C$
5.	$f(b) = A$	是	
6.	$f(x)$ 在 $x = b$ 连续	非	$C = \lim_{x \rightarrow b} f(x) \neq f(b) = A$
7.	$\lim_{x \rightarrow c} f(x) = B$	非	$\lim_{x \rightarrow c} f(x)$ 不存在
8.	$\lim_{x \rightarrow c} f(x) = C$	非	$\lim_{x \rightarrow c} f(x)$ 不存在
9.	$f(x)$ 在 $x = c$ 不连续	是	



c) 若 $f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$,

问 $f(x)$ 在 $x = 1$ 处是否连续。

解: $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

$$= \lim_{x \rightarrow 1} \frac{(x+1)\cancel{(x-1)}}{\cancel{x-1}}$$

$$= 1 + 1$$

$$= 2$$

$$f(1) = 1$$

$$\because \lim_{x \rightarrow 1} f(x) \neq f(1)$$

$\therefore f(x)$ 在 $x = 1$ 处不连续。

思考:

$f(x)$ 为分段函数,
为什么求 $\lim_{x \rightarrow 1} f(x)$ 时候,
不需要个别求 $\lim_{x \rightarrow 1^-} f(x)$ 、 $\lim_{x \rightarrow 1^+} f(x)$?

因为不管是 $x \rightarrow 1^-$ 或 $x \rightarrow 1^+$, $f(x)$ 都是等于 $\frac{x^2 - 1}{x - 1}$

函数连续存在的条件:

i) $\lim_{x \rightarrow x_0} f(x) = A$

ii) $f(x_0) = A$

iii) $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

则 $f(x)$ 在 $x = x_0$ 处连续

d) 若 $f(x) = \begin{cases} 3x-1 & , x < 0 \\ 2x+1 & , x = 0, \\ x^2-1 & , x > 0 \end{cases}$

问 $f(x)$ 在 $x=0$ 及 $x=-2$ 处是否连续。

解：

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} (3x-1) & \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} (x^2-1) \\ &= -1 & &= -1 \end{aligned}$$

$$\because \lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

$$\therefore \lim_{x \rightarrow 0} f(x) \neq -1$$

$$f(0) = 2(0) + 1 = 1$$

$$\because \lim_{x \rightarrow 0} f(x) \neq f(0)$$

$\therefore f(x)$ 在 $x=0$ 不处连续。

思考：

“极限存在”则一定“连续”？

“连续”则一定“极限存在”？

函数连续存在的条件：

$$i) \lim_{x \rightarrow x_0} f(x) = A$$

$$ii) f(x_0) = A$$

$$iii) \lim_{x \rightarrow x_0} f(x) = f(x_0)$$

则 $f(x)$ 在 $x = x_0$ 处连续

$$\begin{aligned} \lim_{x \rightarrow -2} f(x) &= 3(-2) - 1 \\ &= -7 \end{aligned}$$

$$\begin{aligned} f(-2) &= 3(-2) - 1 \\ &= -7 \end{aligned}$$

$$\because \lim_{x \rightarrow -2} f(x) = f(-2)$$

$\therefore f(x)$ 在 $x = -2$ 处连续。

随堂练习 (快问快答)

判断下列函数在 $x = a$ 处是否连续

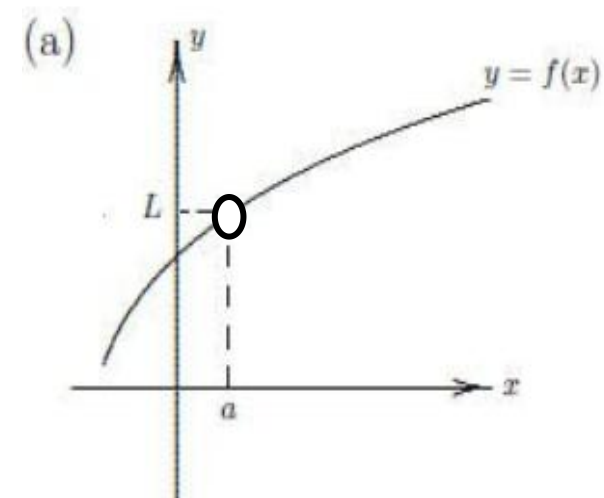
函数连续存在的条件:

i) $\lim_{x \rightarrow x_0} f(x) = A$

ii) $f(x_0) = A$

iii) $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

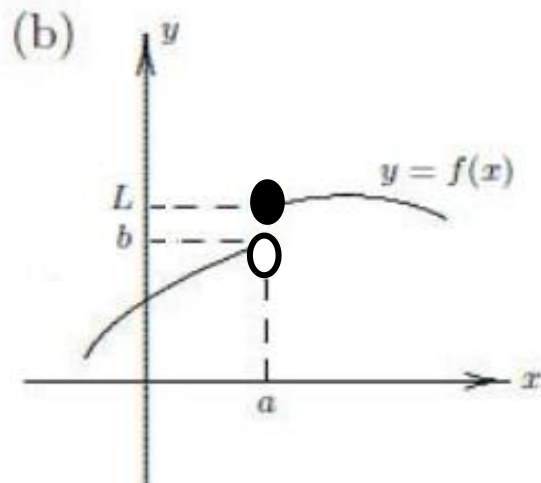
则 $f(x)$ 在 $x = x_0$ 处连续



$$\lim_{x \rightarrow a} f(x) = L$$

$\therefore f(a)$ 无意义

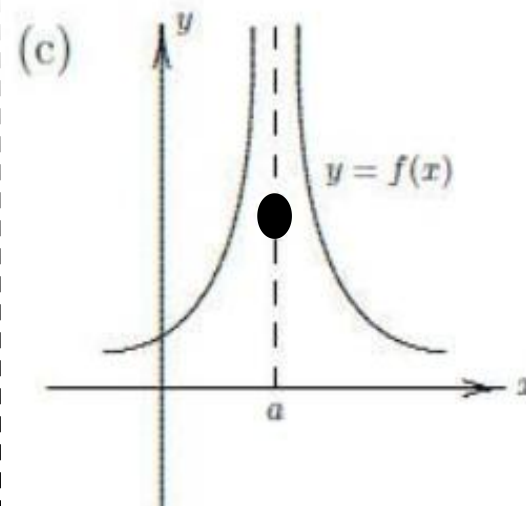
$\therefore f(x)$ 在 $x = a$ 处不连续



$$\therefore \lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$$

$\therefore \lim_{x \rightarrow a} f(x)$ 不存在

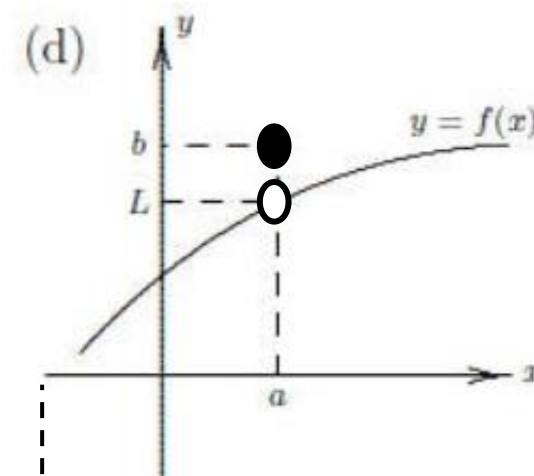
$\therefore f(x)$ 在 $x = a$ 处不连续



$$\therefore \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = \infty$$

$\therefore \lim_{x \rightarrow a} f(x)$ 不存在

$\therefore f(x)$ 在 $x = a$ 处不连续



$$\lim_{x \rightarrow a} f(x) = L$$

$$f(a) = b$$

$$\therefore \lim_{x \rightarrow a} f(x) \neq f(a)$$

$\therefore f(x)$ 在 $x = a$ 处不连续

$$f(x) = \begin{cases} -2x^2 & , \quad x \geq 2 \\ 2x+2 & , \quad 0 \leq x < 2, \\ x-a & , \quad x < 0 \end{cases}$$

i) 已知 $f(x)$ 在 $x=0$ 处连续, 求 a 的值。

$\because f(x)$ 在 $x=0$ 处连续

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{x \rightarrow 0^-} (x-a) = \lim_{x \rightarrow 0^+} (2x+2)$$

$$-a = 2$$

$$a = -2$$

ii) 判断 $f(x)$ 在 $x=2$ 处是否连续?

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (2x+2) = 2(2)+2 = 6$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (-2x^2) = -2(2)^2 = -5$$

$$\because \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

$\therefore f(x)$ 在 $x=2$ 处不连续

练习21.4 -6,9,10,13